16	**Ans: B** Using first law of thermodynamics $\Delta U = \frac{3}{2} nR\Delta T = Q + W_{on}$ $\frac{3}{2} nR\Delta T_1 = 1000 + (-500) = 500$ $\frac{3}{2} nR\Delta T_2 = 2000 + (-500) = 1500$ $\frac{\Delta T_1}{\Delta T_2} = 0.33$